

Implicit differentiation and constrained optimization

When we solve optimization problems for $\mathbf{x}^*$ we oftentimes want to see how $\mathbf{x}^*$ varies as a function of other parameters of interest (*comparative statics*)

Sometimes it's not convenient to solve for $\mathbf{x}^*$ explicitly as a function of those other parameters

Other times we want to derive general results that are independent of functional forms

Example: Suppose that we want to choose a level of effort $x \geq 0$ to maximize utility conditional on difficulty of work $w \geq 1$

Suppose the utility function is $U(x) = 100x - w^x - x^2$

Note that $\frac{d}{dx}a^x = a^x \log a$

This function is concave (why?) so we know that x^* solves $U_x = 0$, or:

$$100 - w^x \log w - 2x = 0$$

We want to see how our optimizing x^* varies as we change w ;
however we can't easily solve the above equation for x^* in terms of w

Implicit differentiation

Let F be a smooth function with $F(x, y) = 0$; we can think of this equation as characterizing y as an implicit function of x that solves $F(x, y) = 0$ if and only if $y = f(x)$

The rule of implicit differentiation: If $F_y \neq 0$ then

$$f'(x) = \frac{dy}{dx} = -\frac{F_x}{F_y}$$

To return to our original problem, x^* solves:

$$F(x, w) = 100 - w^x \log w - 2x = 0$$

$$\begin{aligned} F_w &= -xw^{x-1} \log w - \frac{w^x}{w} \\ &= -w^{x-1}(x \log w + 1) \end{aligned}$$

$$\begin{aligned} F_x &= -\log w * w^x \log w - 2 \\ &= -(2 + w^x (\log w)^2) \end{aligned}$$

$$\frac{dx^*}{dw} = -\frac{F_w}{F_x} = -\frac{w^{x-1}(x \log w + 1)}{2 + w^x (\log w)^2}$$

What can we do with this information?

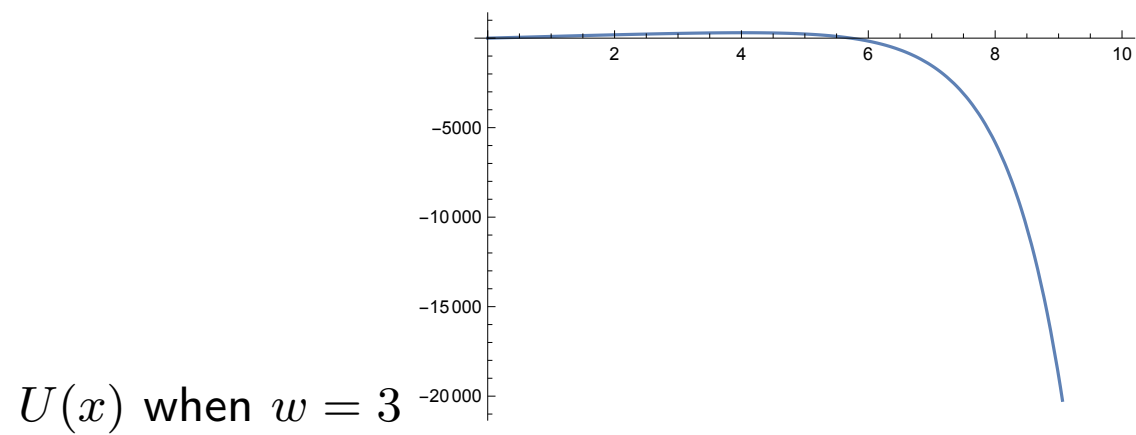
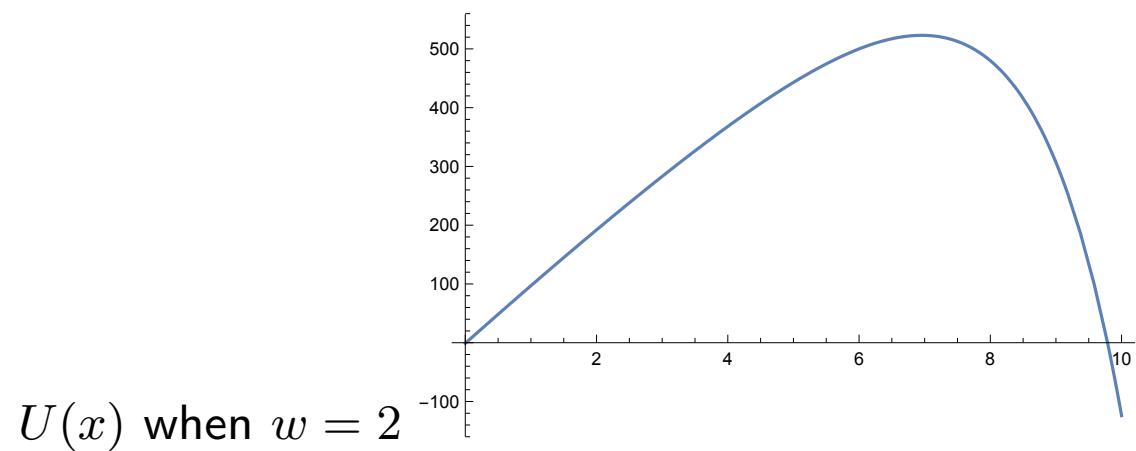
We know that:

$$100 - w^{x^*} \log w - 2x^* = 0$$

and that:

$$\frac{dx^*}{dw} = - \left(\frac{w^{x^*-1} (x^* \log w + 1)}{w^{x^*} (\log w)^2 + 2} \right)$$

- The numerator of $(\cdot)$ is positive when $w^{x^*-1} (x^* \log w + 1) > 0$
 - Or when $\log w > -\frac{1}{x^*}$
 - Since $x \geq 0$, this holds when $w \geq 1$
- The denominator of $(\cdot)$ is *always* positive
- We know that when $w \geq 1$, x^* is decreasing in w
 - When $w = 1$ then $x^* = 50$; when $w \geq e^{100}$ then $x^* = 0$



A last example

Let a firm's revenue be $R(x)$, where x is the quantity of a good produced and sold

The firm can hire labor at a wage w per hour, and $N(x)$ hours are required to produce x units of output; profit is then

$$R(x) - wN(x)$$

The firm wants to choose x^* to maximize profit

- Assume that $N'(x) > 0$ (more hours required to produce more x)
- Assume $R''(x) - wN''(x) < 0$
(profit function strictly concave; what does this imply about N and R ?)

Suppose we want to know how the optimum x^* changes as hourly wage w increases; x^* solves:

$$F(x^*, w) = R'(x^*) - wN'(x^*) = 0$$

Implicit differentiation says that

$$\frac{dx^*}{dw} = -\frac{-N'(x^*)}{R''(x^*) - wN''(x^*)}$$

Since $N'(x) > 0$ and $R''(x) - wN''(x) < 0$ this is negative; thus, x^* is decreasing in w

Optimizing with constraints

We wish to maximize the function $f(x, y)$ subject to the constraint that $g(x, y) = 0$

With two unknowns (x, y) we could solve for $g(x, y) = 0$ in terms of y , substitute back into $f(x, y)$, and then optimize

Example: Maximize $U(x, y) = x^{\frac{1}{3}}y^{\frac{2}{3}}$ subject to $W = 5x + 4y$

This could be reexpressed as “Maximize $U(x, \frac{W-5x}{4})$ ”

This type of substitution isn't always feasible, particularly with more than two variables

The Lagrangian

We want to maximize $f(x, y)$ subject to $g(x, y) = 0$

We start by defining the *Lagrangian* L as a function of three variables: x, y , and λ (the Lagrange multiplier):

$$L(x, y, \lambda) = f(x, y) - \lambda g(x, y)$$

(Given a few assumptions on f and g) any solution to the constrained optimization problem is a critical point of the Lagrangian, and so solves

$$\frac{\partial L}{\partial x} = 0 \quad \frac{\partial L}{\partial y} = 0 \quad \frac{\partial L}{\partial \lambda} = 0$$

The steps

1. Introduce λ and form the Lagrangian L
2. Look at the critical points of L : a critical point (in x and y) will be a constrained maximum of the original problem
3. If there are multiple critical points, examine them by hand to figure out which type of points they are (using a *bordered Hessian*—a topic we haven't covered—or a concavity argument, or some other ad hoc method like a diagram)

Example: Maximize $f(x, y) = xy$ subject to $5x + 2y = 10$

$$L(x, y, \lambda) = xy - \lambda(5x + 2y - 10)$$

$$\nabla L = \begin{bmatrix} y - 5\lambda \\ x - 2\lambda \\ -5x - 2y + 10 \end{bmatrix} = \mathbf{0}$$

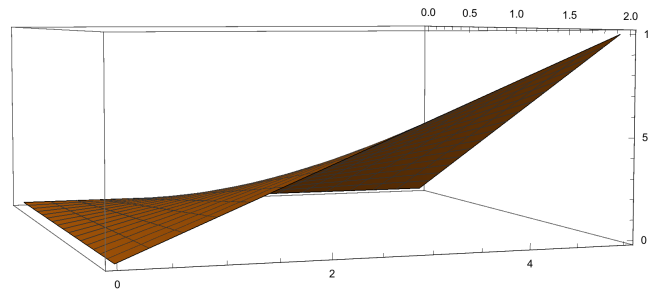
Solving for x, y, λ : $y = 5\lambda$, $x = 2\lambda$,

$$10\lambda + 10\lambda - 10 = 0$$

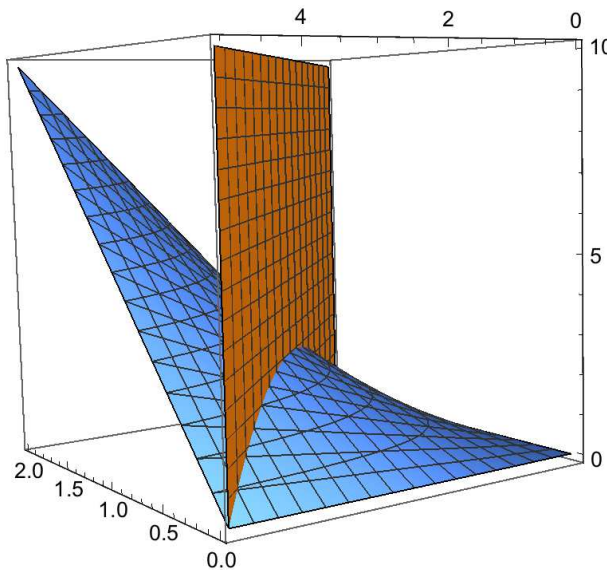
and so $\lambda = \frac{1}{2}$ and $x = 1$ and $y = \frac{5}{2}$

We can plot to confirm $x = 1, y = \frac{2}{5}$ is a constrained (global) max

The function $f(x, y) = xy$



The function $f(x, y) = xy$ and the constraint $y = \frac{10-5x}{2}$



The method of Lagrange multipliers will find all constrained maxima *and* minima, so you need to be careful to check your solutions
(Question: Why didn't our last example find a constrained *minimum*?)

Example: Maximize $5x + y$ subject to $x^2 + y^2 = 10$

$$L(x, y, \lambda) = 5x + y - \lambda(x^2 + y^2 - 10)$$

$$\nabla L = \begin{bmatrix} 5 - 2\lambda x \\ 1 - 2\lambda y \\ -x^2 - y^2 + 10 \end{bmatrix} = \mathbf{0}$$

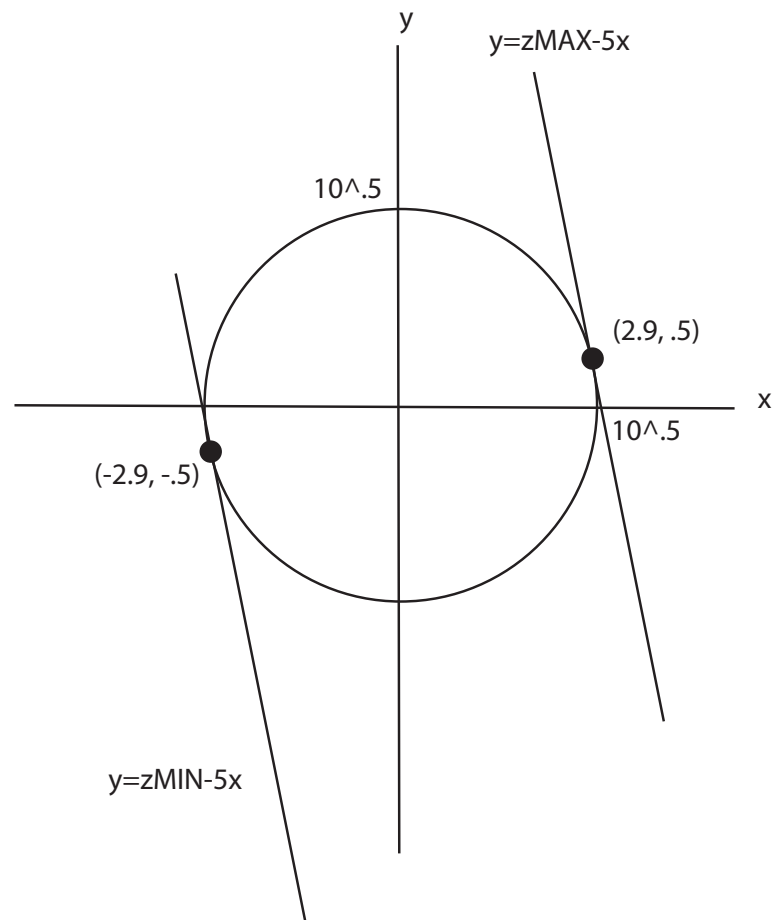
Solving for x, y, λ : $x = \frac{5}{2\lambda}$, $y = \frac{1}{2\lambda}$,

$$\frac{25}{4\lambda^2} + \frac{1}{4\lambda^2} - 10 = 0$$

and so $\lambda = \pm 0.851$ and $(x, y) = (2.93, 0.58)$ and $(-2.93, -0.58)$

Maximize $5x + y$ subject to $x^2 + y^2 = 10$

$(x, y) = (2.93, 0.58)$ is our max $(-2.93, -0.58)$ is our min



Warnings

- If your constraint isn't really a constraint (e.g. our first “minimizing” xy example), L won't find it
- Method provides a *necessary* but not *sufficient* condition for optima
- A constrained optimum will be a critical point of the Lagrangian, but *not necessarily a local maximum of the Lagrangian* (i.e. it could be a saddle of L)
- With one constraint, the gradient of the constraint at an optimum can't equal $\mathbf{0}$
- (For multiple constraints): The gradients of the constraints at an optimum must be linearly independent! You can't have more constraints than unknowns

The meaning of λ

Consider a slightly different problem where we maximize $f(x, y)$ subject to $g(x, y) = b$, with b a constant

The maximum value that f can attain will generally depend on b ; call this maximum value $v(b)$

If (x^*, y^*) is a constrained maximum of f with corresponding λ , then $\lambda = v'(b)$

In words, λ is the rate of change of the quantity being optimized as a function of the constraint parameter b

More variables and more constraints

If we have more than one constraint we create our Lagrangian with a different multiplier for each constraint (though we can't have more constraints than variables)

If we have more than two variables we simply take more first-order conditions to find our critical points

- Potential problem: It's harder to characterize our critical points because we can't draw them out! We generally get around this with concavity / convexity assumptions

Nice things

Suppose that $f(\mathbf{x})$ is a concave function, that $g(\mathbf{x}), h(\mathbf{x}), \dots$ are linear, and that we want to maximize f subject to $g, h, \dots$

Then if $L = f - \lambda(g) - \mu(h) - \dots$ has a critical point at $\mathbf{x}^*$, f attains a constrained (global) maximum at $\mathbf{x}^*$

This extends in the obvious way to convex f and global minima

$\Rightarrow$ Convexity or concavity of the objective function along with linear constraints yields unambiguous critical points

A final example: Maximize $-x^2 - 2y^2 - 3z^2$ subject to $x + y = 5$ and $2x + y + z = 10$

$$L(x, y, z, \lambda, \mu) = -x^2 - 2y^2 - 3z^2 - \lambda(x + y - 5) - \mu(2x + y + z - 10)$$

$$\nabla L = \begin{bmatrix} -2x - \lambda - 2\mu \\ -4y - \lambda - \mu \\ -6z - \mu \\ -x - y + 5 \\ -2x - y - z + 10 \end{bmatrix} = \mathbf{0}$$

$$\nabla L = \begin{bmatrix} -2x - \lambda - 2\mu \\ -4y - \lambda - \mu \\ -6z - \mu \\ -x - y + 5 \\ -2x - y - z + 10 \end{bmatrix} = \mathbf{0}$$

Solving (5 linear equations in 5 unknowns; I used Mathematica) we get

$$x = \frac{25}{6}, y = \frac{5}{6}, z = \frac{5}{6}, \lambda = \frac{5}{3}, \mu = -5$$

Since our objective function is concave and our constraints are linear we know that $x^* = \frac{25}{6}, y^* = \frac{5}{6}, z^* = \frac{5}{6}$ is a constrained optimum